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Two-Step Autologous Grafting Using HYAFF Scaffolds in Treating Difficult Diabetic Foot Ulcers: Results of A Multicenter, Randomized Controlled Clinical Trial With Long-Term Follow-up

Luigi Uccioli, MD[1], Laura Giurato, MD[1], Valeria Ruotolo, MD[1], Adolfo Ciavarella, MD[2], Michele S. Grimaldi, MD[2], Alberto Piaggesi, MD[3], Ilaria Teobaldi, MD[3], Lucia Ricci, MD[4], Luciano Scionti, MD[5], Cristiana Vermigli, MD[5], Roberto Seguro, MD[6], Lorena Mancini, MD[7], and Giovanni Ghirlanda, MD[7]

Abstract

This study evaluated the efficacy and tolerability of an autologous tissue-engineered graft—a 2-step HYAFF autograft—in the treatment of diabetic foot ulcers compared with standard care. In all, 180 patients with dorsal or plantar diabetic foot ulcers (unhealed for ≥1 month) were randomized to receive Hyalograft-3D autograft first and then Laserskin autograft after 2 weeks (n = 90; treatment group) or nonadherent paraffin gauze (n = 90; control group). Efficacy and adverse events were assessed weekly for 12 weeks, at 20 weeks, and at 18 months. The primary efficacy outcome was complete ulcer healing at 12 weeks. Wound debridement, adequate pressure relief, and infection control were provided to both groups. At 12 weeks, complete ulcer healing was similar in both groups (24% of treated vs 21% controls). A 50% reduction in ulcer area was achieved significantly faster in the treatment group (mean 40 vs 50 days; $P = .018$). Weekly percentage ulcer reduction was consistently higher in the treatment group. At 20 weeks, ulcer healing was achieved in 50% of the treated group as compared with 43% of controls. Dorsal ulcers had a 2.17-fold better chance of wound healing per unit time following autograft treatment ($P = .047$). In a subgroup with hard-to-heal ulcers, there was a 3.65-fold better chance of wound healing following autograft treatment of dorsal ulcers ($P = .035$). Adverse events were similar in both groups. The study results demonstrated the potential of this bioengineered substitutes to manage hard-to-heal dorsal foot ulcers.

Keywords

randomized controlled trial, diabetic foot ulcers, HYAFF, tissue engineering, ulcer healing

Diabetic foot ulcers represent a major problem that considerably impairs quality of life and may result in amputation. Peripheral neuropathy and vascular disease are considered to be the major factors leading to chronic foot ulcerations.[1,2] Additionally, in diabetic patients, functional changes in microcirculation and abnormalities in the expression of growth factors involved in the healing process may contribute both to ulcer development and failure to heal.[3-5]

Current management of diabetic foot ulcers includes the restoration of an adequate vascular supply, off-loading, and appropriate local ulcer treatment.[1,6] In recent years, several

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new wound therapies have been developed based on advancements in the understanding of wound physiology and the healing process.[7,8] These include treatments based on living skin equivalents.[9]

These novel treatments include the use of grafts obtained from cultured expanded autologous fibroblasts and keratinocytes grown on scaffolds composed of the benzyl ester of hyaluronic acid, HYAFF-11.[10-12] These dermal and epidermal tissue-engineered autografts (Hyalograft 3D autograft and Laserskin autograft, respectively; Anika Therapeutics srl, Abano Terme, Italy) have been used to treat chronic ulcers with encouraging results on the evolution of the wound bed and on healing times.[13-16]

A randomized, controlled clinical trial conducted by Caravaggi et al[16] compared treatment with Hyalograft-3D autograft followed by Laserskin autograft with standard care in 82 patients with diabetic foot ulcers. The study demonstrated the efficacy of the treatment in this setting with statistically significant differences compared with standard care to treat patients with diabetes and dorsal ulcers.

The authors of this report set up a larger-scale study with long-term follow-up in patients with diabetic foot ulcers. The study objective was to evaluate the efficacy and tolerability of this 2-step autograft treatment compared with standard treatment.

Methods and Materials

This was an open, randomized, controlled study conducted in 7 specialized diabetic foot centers in Italy between September 1999 and January 2006. The clinical trial was approved by the independent ethics committee of each center and conducted according to the Declaration of Helsinki. All patients included were outpatients and gave written informed consent before inclusion in the study.

The main inclusion criteria were type 1 or 2 diabetes mellitus, an ulcer ≥2 cm^2 on the plantar or plantar-marginal surface (excluding calcaneum) or dorsum of the foot without signs of healing for 1 month, Wagner score 1 to 2, transcutaneous partial pressure of oxygen ($TcPO_2$) ≥20 mm Hg, and ankle brachial pressure index (ABPI) ≥0.5. The main exclusion criteria were ulcers with clinical infection, osteomyelitis, inability to tolerate off-loading for pressure relief, and peripheral revascularization <30 days before enrolment.

At the first visit, eligible patients were randomized in a 1:1 ratio, using a computer-generated method, in a block size of 4 and stratified by center. For randomization, each site used sealed envelopes opened in numerical order. A skin biopsy (1-2 cm^2) was taken from each patient randomized to the treatment group and sent to the laboratory of Anika Therapeutics srl (Abano Terme, Italy). After a 2-week run-in period with nonadherent paraffin gauze only, patients with an ulcer area ≥1 cm^2 received their randomized treatment at baseline visits.

All patients had weekly follow-up visits for 12 weeks with a detailed description and photograph of the wound bed and ulcer area tracing measured later by computerized morphometry. All patients received debridement, adequate pressure relief, and treatment of infection, as required by current international guidelines.

The primary efficacy end point was complete ulcer healing at 12 weeks, as demonstrated by complete reepithelialization without exudates and eschar. An investigator-driven assessment was also undertaken to assess ulcer healing at 20 weeks. Secondary efficacy end points were the evaluation of ulcer healing time, the time to achieve 50% of ulcer area reduction, and the weekly percentage of ulcer area reduction. Tolerability was assessed by the collection of adverse events and clinical laboratory results throughout the trial.

Autologous Skin Graft Treatment

Skin cells from biopsy were cultured on the appropriate scaffold, as previously described.[12] At baseline visits, patients received the dermal tissue-engineered autografts (Hyalograft 3D autograft); the graft was covered with nonadherent paraffin gauze and a secondary bandage of sterile cotton pads and gauze. The secondary bandaging could be changed after 5 days, but the nonadherent gauze had to be left for at least 7 days. Approximately 2 weeks later, the ulcer received the epidermal tissue-engineered autograft (Laserskin autograft) covered and dressed in an identical manner. Based on the clinician's judgment, a second Hyalograft-3D autograft or Laserskin autograft application was permitted.

Control Treatment

Control group treatment was nonadherent paraffin gauze (Jelonet; Smith & Nephew, Hull, UK) covered with a secondary dressing of sterile cotton pads and gauze, which could be changed daily or less frequently, depending on the state of the wound bed.[8,16]

Pressure Relief Devices and Footwear

All patients with a plantar ulcer received pressure relief of the affected limb according to the standard practice of the diabetic foot center; those with a dorsal ulcer received therapeutic shoes with a rigid insole. Compliance with off-loading was assessed at each control visit by the clinician.

Statistical Analysis

The original protocol planned 80 for participants, increased by protocol amendment to 200 participants to provide a

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Table 1. Baseline Clinical Characteristics[a]

	Total Population	Dorsal Ulcers	Plantar Ulcers
Treatment group (n)	80	25[b]	52[b]
Diabetes			
Type 1	11 (14%)	1 (4.0%)	10 (19.2%)
Type 2	68 (85%)	24 (96%)	41 (78%)
$TcPO_2$ (mm Hg)	36.50 (10-135)	35.75 (20-135)	37.00 (20-90)
Ankle brachial index	0.9 ± 0.2	0.90 ± 0.21	0.91 ± 0.19
Area of ulcer (cm^2)	8.8 ± 9.4[c]	7.35 ± 5.71	9.04 ± 10.07
Duration of ulcer (months)	7.4 ± 6.6	6.82 ± 5.09	7.44 ± 6.88
Control group (n)	80	30	50
Diabetes			
Type 1	6 (8%)	1 (3.3%)	5 (10%)
Type 2	74 (92%)	29 (96.7%)	45 (90%)
$TcPO_2$ (mm Hg)	36 (20-99)	37 (21-99)	33.5 (20-85)
Ankle brachial index	0.9 ± 0.7	0.99 ± 1.11	0.92 ± 0.21
Area of ulcer (cm^2)	6.7 ± 7.7[c]	5.53 ± 5.37	7.60 ± 9.26
Duration of ulcer (months)	7.3 ± 7.8	5.43 ± 4.83	8.41 ± 8.91

Abbreviations: $TcPO_2$, transcutaneous partial pressure of oxygen; SD, standard deviation.
[a]Data are n (%), means ± SD, or median (minimum − max) for $TcPO_2$.
[b]The total subgroup population in the treatment group is n = 77 because 3 ulcers were not classified.
[c]$P = .016$.

robust assessment of efficacy. Recruitment was stopped at 180 participants in light of the long duration of the enrolment. Efficacy analysis was performed on the intention-to-treat population, defined as all randomized patients without major violations. Tolerability was assessed on the safety population, corresponding to all randomized patients with at least 1 treatment.

Frequency distribution was used for discrete data, and mean and standard deviation (SD) were used to describe continuous data. The comparison between groups was performed using the χ^2 test (discrete data) or Mann-Whitney and Student's *t* test (continuous data). An analysis of covariance model was used considering the treatment group as an explicative factor and the baseline ulcer area as covariate.

At 20 weeks, the healing probability was estimated with survival function computed with the limit product method and compared with the log-rank test. To adjust for the confounding factors effect, the Cox's proportional hazard model was used, using as regressors' baseline ulcer areas, duration of the ulcer, treatment group, and $TcPO_2$. A further post hoc analysis was conducted on a subgroup with non-healing ulcers, defined as those in which ulcer size fails to reduce by half over the first 4 weeks of treatment.[17] The last-observation-carried-forward method was used to replace the missing data. The statistical analysis was conducted using SAS software, version 8.2 for Windows (SAS Inc, Cary, NC). All analyses were 2-tailed to a significance level of 5%.

Results

Efficacy

A total of 180 patients were screened and randomized (n = 90 per group). Of these, 7 patients had an ulcer area <1 cm^2 after the run-in period and were excluded, and 13 patients did not return to the investigational site after the baseline visit. Thus, 160 patients were included in the intention-to-treat analysis (n = 80 per group).

Relevant clinical characteristics at baseline are shown in Table 1. The 2 study groups were similar with the exception of ulcer area, which was significantly larger in the treatment group ($P = .016$). Mean (±SD) age was 61 (±10) years in the treatment group and 62 (±11) years in the control group. The majority of patients in both the treatment and control groups had plantar ulcers—66% (n = 52) and 61% (n = 50), respectively. $TcPO_2$ was similar in the 2 groups, with 25 patients in the treatment group and 29 in the control group with $TcPO_2 < 30$ mm Hg. The others had all $TcPO_2 < 30$ mm Hg. Two grafts of Hyalograft-3D autograft in the 31% (n = 25) and 2 grafts of Laserskin autograft in the 37.5% size (n = 30) were required, corresponding to a mean number of 1.3 ± 0.47 and 1.5 ± 0.59 grafts, respectively. Patients had good compliance to off-loading, which was received by 47 patients in the control group and 53 in the treated group.

Complete ulcer healing at 12 weeks was achieved in 19 (24%) patients in the treatment group versus 17 (21%) in the control group ($P = .850$, ns). The mean time to complete ulcer healing was 50 and 58 days for the treatment and

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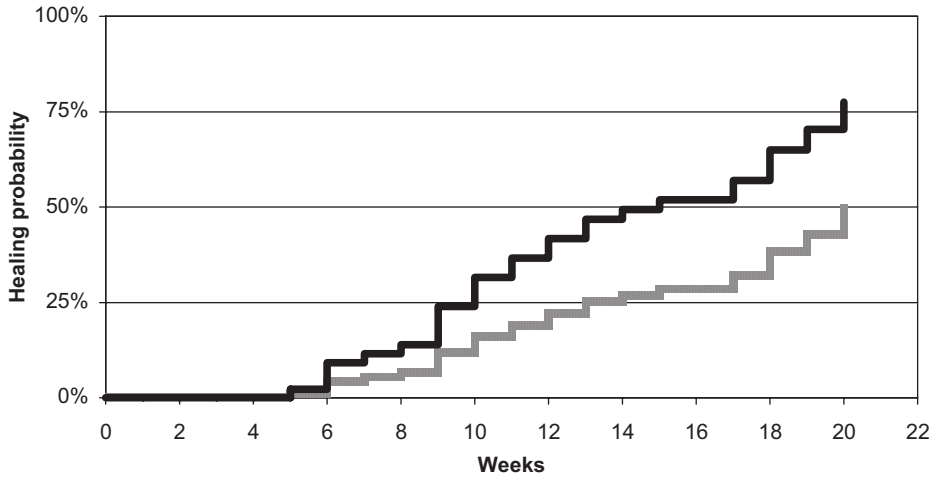

Figure 1. Probability of healing of dorsal ulcers after adjusting for ulcer area and duration: continuous line, treatment group (Hyalograft-3D autograft + Laserskin autograft); dotted line, control group (nonadherent paraffin gauze); hazard ratio = 2.17; $P = .047$

control groups, respectively ($P = .253$, ns). A 50% reduction in ulcer area was achieved significantly faster in the treatment group compared with the control group (mean 40 vs 50 days; $P = .018$). The weekly reduction rate was in favor of the treatment group throughout the study period, and the weekly percentage reduction was significantly higher in the treatment group 2 weeks after baseline (29% vs 14%; $P = .023$).

At 20 weeks, complete ulcer healing was achieved in 50% of the treatment group versus 43% of the control group (log-rank test = 0.344). In the dorsal ulcer subgroup, after adjusting for ulcer area and duration, treatment was found to have a statistically significant effect on the probability of wound healing ($P = .047$). The estimated hazard ratio indicated that an average ulcer treated with Hyalograft-3D autograft and Laserskin autograft had a 2.17-fold better chance for closure per unit time than an ulcer treated with standard care (Figure 1).

The same analysis on the plantar ulcer subgroup indicated that treatment with Hyalograft-3D autograft and Laserskin autograft favored ulcer healing, though this was not statistically significant.

For the subgroup analysis of nonhealing ulcers, we identified a total of 84 nonhealing ulcers in our study population: 28 dorsal, 54 plantar, and 2 with no site recorded. Relevant clinical characteristics are presented in Table 2. In the plantar ulcer group, the treatment did not appear to influence outcome. However, in nonhealing dorsal ulcers, treatment with autologous skin substitutes had a statistically significant beneficial effect on the probability of wound healing ($P = .035$). The estimated hazard ratio indicated that an average ulcer in this particular subpopulation treated with Hyalograft-3D autograft and Laserskin autograft had a 3.65-fold better chance for closure per unit time than that in a control patient.

Table 2. Baseline Clinical Characteristics of Nonhealing Ulcers[a]

	Treatment Group	Control Group
n	37	47
Diabetes		
Type 1	5 (13.5%)	4 (8.5%)
Type 2	32 (86.5%)	43 (91.5%)
$TcPO_2$ (mm Hg)	34.75 (20-90)	36 (21-90)
Ankle brachial index	0.92 ± 0.17	0.89 ± 0.23
Area of ulcer (cm^2)	10.02 ± 10.80	7.84 ± 9.15
Duration of ulcer (months)	6.56 ± 4.97	8.37 ± 9.04

Abbreviations: $TcPO_2$, transcutaneous partial pressure of oxygen; SD, standard deviation.
[a]Data are n (%), means ± SD, or median (min − max) for $TcPO_2$.

Safety

The safety population was composed of 171 patients (n = 84 treated; n = 87 controls). There were no significant differences between the 2 groups in terms of mean change from baseline to 12 weeks for any laboratory parameter. During the 12-week study period, adverse events occurred in 18 (21%) treated patients and 14 (16%) controls. No event was considered related to study treatment. The majority of events (41/46) were mild or moderate. In both groups,

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Table 3. Serious Adverse Events (SAEs)

Center No./ Patient No.	Group	SAE Description	Severity
5/15	Control	Thrombophlebitis	Moderate
5/19	Treated	Amputation	Severe
7/9	Treated	Hematemesis	Severe
2/6	Treated	Amputation	Moderate
3/3	Treated	Myocardial failure	Moderate
6/29	Treated	Joint effusion	Severe
6/19	Treated	Infection	Severe
1/1	Control	Pleural effusion	Moderate
5/25	Treated	Osteomyelitis	Severe

the most frequent adverse event was infection—15.4% (n = 13) in the treatment group and 11.4% (n = 10) in the control group. Overall, 9 serious adverse events occurred (Table 3): 7 events in the treated group and 2 events in the control group ($P = .2621$, ns); none was considered related to study treatment.

A total of 103 patients attended the 18-month follow-up visit (51 and 52 patients in the treated and control group, respectively); 9 adverse events were reported by 1 (2%) treated patient and 8 (15.4%) controls. One of these patients (control group) died. None of the events was considered related to study treatment.

Discussion

In this randomized multicenter controlled clinical trial, the use of a 2-step procedure with autologous skin grafts—the Hyalograft-3D autograft and Laserskin autograft—was compared with standard care in a larger study population than in previous studies and with longer follow-up. Safety and tolerability results were excellent, and no adverse events were related to study treatment. The primary efficacy outcome of complete ulcer healing was not statistically significantly different ($P > .05$, ns) between the groups: at 12 weeks, it was achieved by 24% and 21% in the treated and control groups, respectively; at 20 weeks, it was 50% versus 43% in favor of the treated group, the difference being not statistically significant. The mean time to complete ulcer healing was lower in the treated group (50 days) than in the control group (58 days), though not statistically different. It should be noted that the mean ulcer area at baseline was greater in the treatment than in the control group (mean ulcer of 8.8 cm^2 vs 6.6 cm^2at baseline). In previous studies of diabetic foot ulcers with skin substitutes, the mean ulcer area of the treated group was lower than that in our study population.[18-22] We are aware that ulcer area is commonly recognized as one of the risk factors influencing ulcer healing[23]; in particular, an ulcer area >5 cm^2 has been reported to be predictive of ulcer healing in more than 4 months.[14] It is possible that the initial difference in ulcer size affected the significance levels of our finding.

Conversely, statistically significant differences between tissue-engineered autograft treatment and standard care were demonstrated at 12 weeks for the time to achieve 50% reduction in ulcer area and weekly percentage of ulcer area reduction. Furthermore, considering the dorsal ulcer subgroup at 20 weeks, a positive statistically significant result was obtained on the probability of complete wound healing ($P = .047$) in concord with the results previously obtained by Caravaggi et al[16] within the same subgroup of diabetic foot ulcers, whereas in both clinical trials, the use of autologous grafts in the plantar ulcer subgroup did not show significantly better healing. This supports previous observations[16] that plantar foot ulceration is more sensitive to off-loading and suggests that better-designed studies may be needed to demonstrate treatment efficacy.

The efficacy of Hyalograft-3D autograft and Laserskin autograft treatment was also shown in patients with nonhealing dorsal ulcers. Autologous tissue-engineered skin substitutes had a statistically significant effect on the probability of wound healing ($P = .035$), showing that an average ulcer of this particular subpopulation treated with the Hyalograft-3D autograft and Laserskin autograft had a 3.65-fold better chance for closure per unit time than a patient treated with standard care. This subpopulation had a poor prognosis because they were patients at particular risk of amputation with a median $TcPO_2$ <40 mm Hg, as reported by Faglia et al.[24]

Thus, the present study has its limitations. It was ended prematurely because of the low enrollment with fewer number of randomized patients than initially planned, with larger ulcers at baseline in the treated group, which may have underpowered the trial and included hard-to-heal ulcers in the treated group. Furthermore, the inclusion of noninfected ulcers of Wagner 1-2 may limit the generalization of the results to all patients with foot ulcers. However, patients with infected ulcers become the same as those with uninfected ulcers when wound infection clears up. Therefore, because skin substitutes are considerably expensive technologies, they should not be the first choice of treatment for all patients; in fact, the results of this trial should be considered particularly relevant for patients with dorsal hard-to-heal ulcers in whom other treatments fail or are not applicable.

In conclusion, the results demonstrate the safety and effectiveness of autologous skin substitutes in the hard-to-heal diabetic dorsal foot ulcer population. The results permit the suggestion that such bioengineered substitutes are potentially useful in patients with hard-to-heal diabetic dorsal ulcers, and there is a need for larger studies to clearly demonstrate treatment benefits and how suitable patients may be identified.

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References

1. Rathur HM, Boulton AJ. Pathogenesis of foot ulcers and the need for offloading. *Horm Metab Res.* 2005;37(suppl 1): 61-68.
2. Pecoraro RE, Reiber GE, Burgess EM. Pathways to diabetic limb amputation: basis for prevention. *Diabetes Care.* 1990;13:513-521.
3. Falanga V. Wound healing and its impairment in the diabetic foot. *Lancet.* 2005;366:1736-1743.
4. Hehenberger K, Kratz G, Hansson A, Brismar K. Fibroblasts derived from human diabetic wound have a decreased proliferation rate, which is healed by the addition of heparin. *J Dermatol Sci.* 1998;16:144-151.
5. Sternberg M, Cohen-Forterre L, Peyroux J. Connective tissue in diabetes mellitus: biochemical alterations of the intercellular matrix with special reference to proteoglycans, collagens and basement membranes. *Diabete Metab.* 1985;11:27-50.
6. Saap LJ, Falanga V. Debridement performance index and its correlation with complete closure of diabetic foot ulcers. *Wound Repair Regen.* 2002;10:354-359.
7. Claxton MJ, Armstrong DG, Boulton JM. Healing the diabetic wound and keeping it healed: modalities for the early 21st Century. *Curr Diab Rep.* 2002;2:510-518.
8. Margolis DJ, Kantor J, Berlin JA. Healing of diabetic neuropathic foot ulcers receiving standard treatment: a meta-analysis. *Diabetes Care.* 1999;22:692-695.
9. Kirsner RS, Falanga V, Kerdel FA, Katz MH, Eaglstein WH. Skin grafts as pharmacological agents: pre-wounding of the donor site. *Br J Dermatol.* 1996;135:292-296.
10. Campoccia D, Doherty P, Radice M. Semi-synthetic resorbable materials from hyaluronan derivatization. *Biomaterials.* 1998;19:2101-2127.
11. Galassi G, Brun P, Radice M, et al. In vitro reconstructed dermis implanted in human wounds: degradation studies of the HA-based supporting scaffold. *Biomaterials.* 2000;21: 2181-2191.
12. Harris PA, di Francesco F, Barisoni D, Leigh IM, Navsaria HA. Use of hyaluronic acid and cultured autologous keratinocytes and fibroblasts in extensive burns. *Lancet.* 1999;353: 35-36.
13. Cavallini M. Autologous fibroblasts to treat deep and complicated leg ulcers in diabetic patients. *Wound Repair Regen.* 2007;15:35-38.
14. Uccioli L; TissueTech Autograft System Italian Diabetic Foot Ulcer Study Group. Clinical results related to the use of the TissueTech Autograft System in the treatment of diabetic foot ulceration. *Wounds.* 2003;15:279-288.
15. Faglia E, Mantero M, Gino M, Quarantiello A, Signorini M. A combined conservative approach in the treatment of a severe Achilles tendon region ulcer in a diabetic patient. *Wounds.* 1999;11:105-109.
16. Caravaggi C, De Giglio R, Pritelli C, et al. HYAFF-11 based autologous dermal and epidermal grafts in the treatment of noninfected diabetic plantar and dorsal foot ulcers: a prospective, multicenter, controlled, randomized clinical trial. *Diabetes Care.* 2003;26:2853-2859.
17. Sheehan P, Jones P, Casella A, Giurini JM, Veves A. Percent change in wound area of diabetic foot ulcer over a 4-week period is a robust predictor of complete healing in a 12-week prospective trial. *Diabetes Care.* 2003;26:1879-1882.
18. Martson WA, Hanft J, Norwood P, Pollak R. The efficacy and safety of Dermagraft in improving the healing of chronic diabetic foot ulcers. *Diabetes Care.* 2003;26:1701-1705.
19. Hunter HS, Chen J, King LE. Graftskin treatment of difficult to heal diabetic foot ulcers: one centre's experience. *Dermatol Surg.* 2002;28:698-703.
20. Veves A, Falanga V, Armstrong D, Sabolinski ML. Graftskin, a human skin equivalent, is effective in the management of noninfected neuropathic diabetic foot ulcer. *Diabetes Care.* 2001;24:290-295.
21. Gentzkow GD, Jensen JL, Pollak RA, Kroeker RA, Lerner JM, Iwasaki SD. Improved healing of diabetic foot ulcers after grafting with a living human dermal replacement. *Wounds.* 1999;11:77-84.
22. Pollak RA, Edington H, Jensen JL, Kroeker RO, Gentzkow GD; the Dermagraft Diabetic Ulcer Study Group. A human dermal replacement for the treatment of diabetic foot ulcer. *Wounds.* 1997;9:175-183.
23. Oyibo SO, Jude EB, Tarawneh I, et al. The effects of ulcer size and site, patient's age, sex and type and duration of diabetes on the outcome of diabetic foot ulcers. *Diabet Med.* 2001;18:133-138.
24. Faglia E, Clerici G, Caminiti M, Quarantiello A, Curci V, Morabito A. Predictive values of transcutaneous oxygen tension for above-the-ankle amputation in diabetic patients with critical limb ischemia. *Eur J Vasc Endovasc Surg.* 2007; 33:731-736.

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